2. Wednesday, March 27. Compound Statements and logical operators (1.1, 2.1)

Last Time: · Defined statements

· Established that we need to learn how to prove.

⟶ Need to explore!

Definition 1: An integer larger than 1 is said to be prime if it cannot be written as a product of two smaller positive integers.

Next: study factorizations/primeness of $2^n - 1$

n	$2^n - 1$
2	3
3	7
4	15
5	31
6	63
7	127
8	255
9	$511 = 7 \cdot 73$
10	$1023 = 3 \cdot 341$
11	$2047 = 23 \cdot 89$

Conjecture A:

If n is prime, then $2^n - 1$ is prime.

This is false, 11 is prime but $2^{11} - 1$ is not

Conjecture B:

If n is not prime, then $2^n - 1$ is not prime

This one seems to be true, but can't decide that from this (or any other finite) table.

We need a proof!

Compound Statements: Are formed combining statements with various logical operators

Example: "Orcas are mammals and glaciers can move"

Definition 2: Suppose P and Q are statements

- The conjuction of P and Q is the statement "P and Q". denoted $P \wedge Q$. (\land)
"P and Q" true if both P and Q are true, false otherwise.
- The disjunction of P and Q is the statement "P or Q" denoted $P \vee Q$. (\lor)
"P or Q" is true if P is true or Q is true (or both are), it's false if both P and Q are false.
- The negation of P is the statement "not P". denoted $\neg P$ (\lnot)
"not P" is true if P is false, and false if P is true.
- The conditional is the statement "if P then Q" denoted $P \to Q$ (denoted \to)
"if P then Q" is false when P is true and Q is false. It's true otherwise.
e.g. P: "it is raining", Q: "I bring an umbrella"
- The biconditional is the statement "P if and only if Q". denoted $P \leftrightarrow Q$ (\leftrightarrow)
"P if and only if Q" is true when P and Q have the same truth value, false otherwise.

Remarks

- In math, "or" is inclusive unless stated.
  This means that "P or Q" is true if both P and Q are true.

- In natural language, there are many ways to negate a statement. Use one that's natural.
  For example, for P: "3 divides 6", we get ¬P: "not 3 divides 6" Bad :(
  : "3 does not divide 6" Good :)

- In a conditional $P \to Q$, Q → conclusion/predicate; P → hypothesis/antecedent

Truth table:

P	Q	P → Q	
T	T	T	} Make sense
T	F	F	
F	T	T	} If hypothesis is false, conditional is true
F	F	T	

Truth Table: contains truth values of a compound statement depending on the truth values of the statements used to form it.

Basic Truth Tables:

P	Q	$P \wedge Q$	$P \vee Q$	$P \rightarrow Q$	$P \leftrightarrow Q$
T	T	T	T	T	T
T	F	F	T	F	F
F	T	F	T	T	F
F	F	F	F	T	T

Example: Suppose we have statements P, Q and R. Write the truth table for $\neg Q \rightarrow (P \vee R)$

P	Q	R	$\neg Q$	$P \vee R$	$\neg Q \rightarrow (P \vee R)$
T	T	T	F	T	T
T	T	F	F	T	T
T	F	T	T	T	T
T	F	F	T	T	T
F	T	T	F	T	T
F	T	F	F	F	T
F	F	T	T	T	T
F	F	F	T	F	F

False only when $\neg Q$ True and $P \vee R$ False

More on conditionals:

1) Conditional statements are very common because they encode causation.
2) Several ways of saying the statement $P \longrightarrow Q$ in English. For example:
   - If P, then Q
   - P implies Q
   - Q is true whenever P is true.
   - P is sufficient for Q.
   - Q is necessary for P.
   - P only if Q
   - Q if P

Exercise: Substitute P: "It's raining" and Q: "I bring an umbrella"